

AI-Powered Learning Management System

Week 1 Engineering Design Package

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DOCUMENT 1: PRODUCT REQUIREMENTS DOCUMENT (PRD)

1. Executive Summary

This AI is a next-generation, enterprise-grade Learning Management System (LMS) designed to transform traditional digital courseware into an interactive, AI-assisted learning experience. Traditional LMS platforms serve as passive content repositories, forcing students to navigate static lecture notes and PDFs without immediate academic support outside classroom hours.

This AI solves this challenge by pairing a robust LMS (course authoring, document management, timed assessments, and progress tracking) with an embedded **Contextual AI Tutor**. Unlike generic public AI chatbots that generate unverifiable or out-of-syllabus answers, the LearnPulse AI Tutor uses **Retrieval-Augmented Generation (RAG)** to answer student questions strictly using course materials uploaded by instructors, delivering grounded answers accompanied by exact page citations.

2. Problem Statement

Modern educational institutions face three core operational and pedagogical challenges:

1. **Unverifiable Public AI Hallucinations:** Public AI chatbots answer queries using unvetted internet data. Students cannot rely on raw LLM responses for course-specific examinations or syllabus compliance.
 2. **Faculty Overload & Repetitive Queries:** Instructors spend substantial hours answering routine administrative and foundational content questions already covered in lecture slides.
 3. **Passive Student Engagement:** Static PDF reading yields lower comprehension and higher drop-off rates compared to active, conversational learning.
-

3. Project Vision

To create a secure, verifiable, and intelligent learning ecosystem where every student enjoys 24/7 access to a personalized AI Tutor strictly aligned with their course curriculum, while empowering educators with automated assessment tools and actionable learning analytics.

4. Objectives & Key Results (OKRs)

- **Objective 1: Grounded Academic AI:** Ensure 100% of AI Tutor responses are derived directly from instructor-approved course documents with verifiable inline page citations.
 - **Objective 2: Accelerated Assessment Execution:** Reduce teacher quiz creation and grading overhead by 70% via automated grading engines and intuitive question builders.
 - **Objective 3: Responsive User Experience:** Deliver sub-200ms REST API response times (P_{95}) and sub-1.2s Time-To-First-Token (TTFT) for AI chat streaming.
 - **Objective 4: Data Security & Access Isolation:** Guarantee strict role-based data isolation so students access only enrolled course materials.
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5. Project Scope

LEARNPULSE AI PRODUCT SCOPE		
CORE LMS MODULES	CONTEXTUAL AI MODULES	
- Identity & Role Management	- PDF Ingestion & Text Parsing	
- Subject & Chapter Syllabus Builder	- Passage Chunking & Vector Indexing	
- Rich-Text Notes & PDF Distribution	- Course-Bounded RAG Q&A; Engine	
- Timed Quiz Builder & Auto-Grading	- Inline Page & Passage Citations	
- Student Progress & Class Analytics	- Prompt Injection Guardrails	

7. Functional Requirements

7.1 Authentication & Role Management

- **FR-AUTH-01:** Secure user authentication via email credentials and password validation.
- **FR-AUTH-02:** Stateless session authorization supporting Student, Teacher, and Administrator roles.
- **FR-AUTH-03:** Secure password recovery via time-bound recovery tokens.

7.2 Student Learning Workspace

- **FR-STU-01:** Browse enrolled subjects, sequential chapters, rich-text lecture notes, and attached PDF slides.
- **FR-STU-02:** Launch interactive AI Tutor sessions scoped to a specific subject, chapter, or document.
- **FR-STU-03:** Attempt timed chapter quizzes with automated submission timers and instant score breakdowns.
- **FR-STU-04:** Track personal progress metrics (reading completion rates, quiz grades, AI study queries).

7.3 Teacher Workspace

- **FR-TCH-01:** Create and manage assigned subjects, chapters, notes, and document attachments.
- **FR-TCH-02:** Upload PDF study files, automatically triggering background text parsing and semantic indexing.
- **FR-TCH-03:** Author chapter quizzes with configurable question pools, time limits, and passing thresholds.
- **FR-TCH-04:** View aggregate class analytics, student grade distributions, and high-frequency student AI queries.

7.4 Administrator Workspace

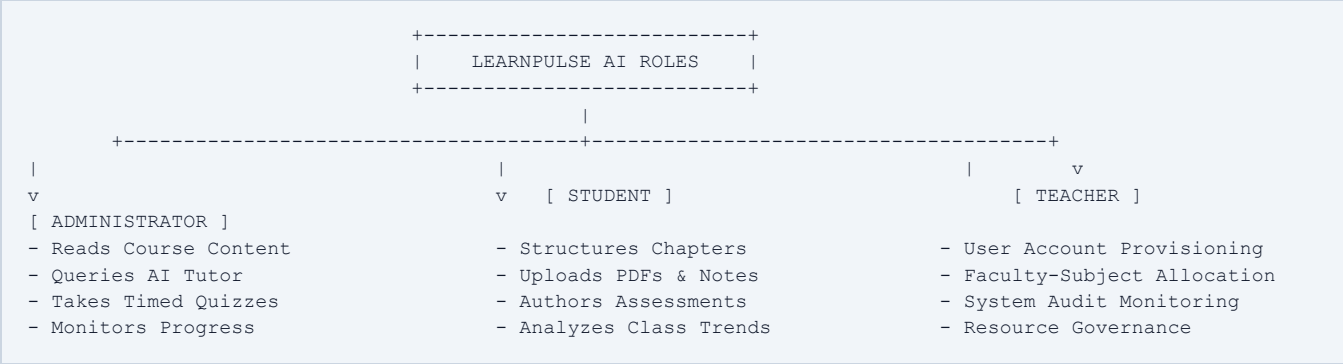
- **FR-ADM-01:** Provision user accounts, edit profiles, toggle account active states, and assign user roles.
- **FR-ADM-02:** Allocate faculty members to specific academic subjects.
- **FR-ADM-03:** Inspect system health metrics, storage utilization, vector search latency, and audit logs.

8. Non-Functional Requirements

Requirement Category	Metric / SLA	Justification
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API Latency	$\$ < 200 \backslash \text{text{ ms}} \$ (\$ P_{\{95\}} \$)$	Ensures smooth dashboard navigation and UI responsiveness
Vector Search Latency	$\$ < 50 \backslash \text{text{ ms}} \$$	Maintains fast document passage retrieval in pgvector
AI Stream Latency	$\$ < 1.2 \backslash \text{text{ s}} \$ \text{ TTFT}$	Provides immediate feedback when streaming AI answers
System Availability	$\$ 99.9 \% \$ \text{ Uptime}$	High availability for student exam windows
Data Integrity	$\$ 100 \% \$ \text{ ACID Compliance}$	Prevents corrupted quiz attempts or orphan vector embeddings
Security Standard	TLS 1.3 & Role Isolation	Protects student records and institutional intellectual property

9. User Roles & Responsibilities



User Story 2: Teacher PDF Upload & Ingestion

- **As a** Course Instructor,
- **I want to** upload lecture PDF slides to a chapter,
- **So that** the system automatically indexes the content for AI study assistance.
- **Acceptance Criteria:**
 - Uploading a valid PDF (<50\text{MB}) displays a "Processing" status badge.
 - Text extraction and semantic vector creation execute asynchronously in the background.
 - Once processed, the document status updates to "Ready," enabling student AI queries against it.

User Story 3: Student Timed Quiz Attempt

- **As a** Student,
- **I want to** attempt a timed chapter quiz,
- **So that** I can test my understanding before final exams.
- **Acceptance Criteria:**
 - Clicking "Start Quiz" initializes a countdown timer based on the quiz's duration limit.
 - If the timer reaches 0:00, the system automatically submits open answers.
 - Upon submission, the student immediately views their final grade percentage and correct option explanations.

11. Success Criteria & Key Performance Indicators (KPIs)

1. **AI Citation Precision:** > 98\% of AI-generated statements correctly map to retrieved document passages.
2. **User Engagement:** Average student study time increases by ≥ 40\% compared to static PDF reading.
3. **Assessment Efficiency:** 100\% of Multiple Choice quizzes are auto-graded instantly upon submission.
4. **Platform Stability:** Zero unexpected database downtimes during peak exam submission windows.

12. Business Value & Impact

- **For Educational Institutions:** Modernizes digital infrastructure, increases student retention, and provides scalable academic support without increasing teaching staff headcount.
- **For Faculty:** Eliminates manual grading of standard quizzes and reduces repetitive administrative inquiries.
- **For Students:** Delivers personalized, 24/7 academic tutoring grounded in verified course materials.

13. Risk Analysis & Mitigation Matrix

Identified Risk	Severity	Likelihood	Risk Mitigation Strategy
AI Prompt Injection Attacks	High	Medium	Sanitize user prompts; enforce strict system prompt guardrails overriding off-topic requests.

Vector Search Latency Spikes	Medium	Low	Scope similarity searches by chapter_id; optimize HNSW vector indexes in pgvector.
Large File Upload Failures	Medium	Medium	Validate file MIME types; enforce 50MB file size limits; process ingestion asynchronously.
Unauthorized Document Sharing	High	Low	Store files privately in cloud storage; issue temporary access links valid for 15 minutes.

14. Week 1 Summary & Next Steps

This Product Requirements Document completes the functional and strategic baseline for **LearnPulse AI**.

Engineering Kickoff Next Steps:

1. Review High-Level Architecture (Document 2) and Database Schemas (Document 3).
2. Establish API contract specifications for frontend and backend integration.
3. Prepare staging environment infrastructure for database and cloud storage initialization.

DOCUMENT 2: HIGH LEVEL DESIGN (HLD)

1. System Architecture Overview

LearnPulse AI adopts a **clean, layered architecture** separating presentation, API security, application services, and stateful storage tiers.

```

graph TD
    subgraph Client_Layer [Presentation Layer]
        SPA[React 18 Single Page Application]
    end

    subgraph Gateway_Layer [Security & Gateway Layer]
        Nginx[Nginx Reverse Proxy / TLS Endpoint]
        Security[Spring Security & JWT Filter Chain]
    end

    subgraph Application_Layer [Application Service Layer - Spring Boot]
        AuthSvc[Auth & Role Service]
        ContentSvc[Subject & Content Service]
        QuizEngine[Quiz & Assessment Engine]
        RAGEngine[RAG & AI Orchestration Service]
        IngestSvc[Document Processing Service]
    end

    subgraph Infrastructure_Layer [Data & AI Infrastructure]
        RelationalDB[(PostgreSQL 16 Operational Store)]
        VectorDB[(pgvector Semantic Vector Store)]
        ObjectStore[(S3 Object Storage - Documents)]
        LLMPProvider[External LLM Service Provider]
    end

    SPA <--> Nginx
    Nginx <--> Security
    Security <--> Application_Layer
    Application_Layer <--> RelationalDB
    IngestSvc <--> ObjectStore
    IngestSvc <--> VectorDB
    RAGEngine <--> VectorDB
    RAGEngine <--> LLMPProvider

```

2. Core System Components

1. **React SPA Client:** Delivers an interactive user workspace for students, teachers, and administrators. Communicates via REST APIs and Server-Sent Events (SSE).
2. **Security Gateway:** Nginx terminates TLS connections, routes API traffic, and enforces gateway rate limits. Spring Security validates access tokens and role privileges.
3. **Spring Boot Core Services:**
 - ***Auth Service*:** Manages user credentials, token issuance, and password security.
 - ***Content Service*:** Manages subjects, chapters, notes, and file metadata.
 - ***Document Ingestion Service*:** Parses uploaded PDFs, extracts text passages, and generates semantic vectors.
 - ***Quiz Engine*:** Controls assessment creation, timers, submission grading, and score persistence.
 - ***RAG Engine*:** Manages vector similarity searches, prompt construction, safety guardrails, and streaming AI responses.
4. **Data Infrastructure:**
 - ***PostgreSQL 16*:** Manages relational data (users, courses, quizzes, attempts).
 - ***pgvector Extension*:** Stores semantic passage vectors natively within PostgreSQL, enabling fast vector searches alongside relational data.
 - ***S3 Object Storage*:** Cloud storage for uploaded PDF study materials.

- ***LLM Provider Gateway***: Interface for vector embedding generation and streaming text completion.

3. Layered Architecture Deep Dive

LEARNPULSE AI LAYER RESPONSIBILITIES		
1. PRESENTATION LAYER	React 18 SPA, UI Components, SSE Streaming Client	
2. SECURITY GATEWAY LAYER	TLS 1.3, Nginx Reverse Proxy, Spring Security Filter	
3. SERVICE APPLICATION	Spring Boot Services, Ingestion Engine, Quiz Engine	
4. DATA STORAGE LAYER	PostgreSQL 16, pgvector Store, S3 Cloud Storage	
5. AI INTEGRATION LAYER	Semantic Embedding Model, Large Language Model	

4. Module Breakdown & Responsibilities

Module	Responsibilities	Dependencies	Data Managed
Authentication	User login, token issuance, session checks	User Service	users, refresh_tokens
Student Portal	Course reading, study tools, progress tracking	Content, Quiz, AI Tutor	student_progress
Teacher Portal	Course creation, PDF uploads, quiz authoring	Content, Ingestion, Analytics	subjects, chapters, notes
Admin Portal	Account provisioning, subject allocations	User Service, Audit Service	users, audit_logs
Content Management	Structuring subjects, chapters, notes, and files	S3 Storage, Ingestion Svc	notes, documents
Document Ingestion	PDF text extraction, passage chunking, vectors	Content, pgvector	document_chunks, embeddings
Quiz Engine	Timed assessments, answer evaluation, scoring	Student, Teacher	quizzes, quiz_attempts
AI Tutor (RAG)	Passage retrieval, prompt assembly, streaming Q&A;	pgvector, LLM Gateway	ai_conversations, messages

5. Component Interaction & Communication Protocols

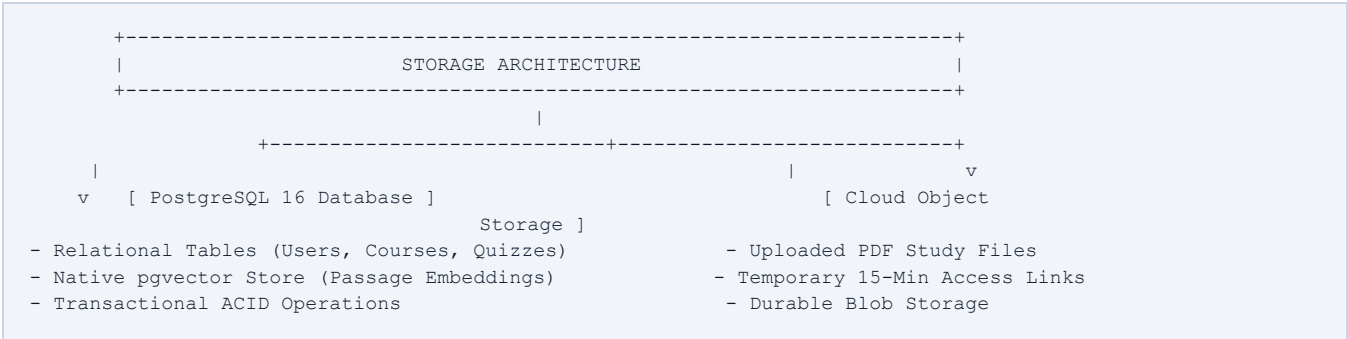
- **Client $\rightarrow$ Gateway**: Secure HTTPS REST endpoints for standard data operations; Server-Sent Events (SSE) for low-latency streaming AI chat completions.

- **Backend Services $\rightarrow$ Database:** PostgreSQL connection pool (HikariCP) handling transactional queries and vector similarity searches.
- **Backend Services $\rightarrow$ Object Storage:** S3 API protocol for secure file uploads and presigned link generation.
- **Backend Services $\rightarrow$ LLM Provider:** Secure HTTPS API requests for embedding generation and completions.

6. Storage Strategy

BEST PRACTICE:

****Unified Database Strategy**:** Using PostgreSQL with `pgvector` allows operational data (users, courses, quizzes) and vector embeddings to reside within the same database engine. This eliminates the need to synchronize separate relational and vector databases.



7. AI Subsystem Integration Overview

The AI integration follows a **Retrieval-Augmented Generation (RAG)** workflow:

1. **Passage Retrieval:** When a student asks a question, the backend queries `pgvector` for the top matching document passages from that chapter.
2. **Context Injection:** Retrieved passages are combined with safety guardrails into a structured system prompt.
3. **Grounded Completion:** The prompt is sent to the LLM, which streams back a response containing explicit page citations.

8. Authentication & Session Flow

```
sequenceDiagram
    actor User as User Browser
    participant Gateway as Gateway / Security Filter
    participant AuthSvc as Auth Service
    participant DB as PostgreSQL Database

    User->>Gateway: POST /api/v1/auth/login {email, password}
    Gateway->>AuthSvc: Forward Credentials
    AuthSvc->>DB: Fetch User Record & Password Hash
    DB-->>AuthSvc: Return User Details
    AuthSvc->>AuthSvc: Verify Password Hash
    AuthSvc->>AuthSvc: Generate Access Token & Refresh Cookie
    AuthSvc-->>User: Return User Profile + Authorization Tokens
    User->>Gateway: Subsequent API Request with Bearer Token
    Gateway->>Gateway: Validate Token & Authorize Request
```

9. Deployment Architecture & Infrastructure Topology

```
graph TB
    subgraph Internet_Edge [Public Internet Edge]
        Users[Web Clients & Mobile Browsers] <--> LoadBalancer[Cloud Load Balancer / Nginx Gateway]
    end

    subgraph Application_Tier [Stateless Application Tier]
        LoadBalancer <--> AppNode1[Spring Boot Application Server 1]
        LoadBalancer <--> AppNode2[Spring Boot Application Server 2]
    end

    subgraph Stateful_Data_Tier [Stateful Data Infrastructure]
        AppNode1 <--> DB[(PostgreSQL 16 Primary + pgvector)]
        AppNode2 <--> DB
        AppNode1 <--> Storage[(S3 Object Storage - Documents)]
        AppNode2 <--> Storage
    end

    subgraph External_AI_Tier [Third-Party AI Services]
        AppNode1 <--> LLM[External LLM Service API]
    end
```

10. High-Level System Sequence Diagram

```
sequenceDiagram
    actor Student
    participant Gateway as API Gateway
    participant RAG as RAG Service
    participant VectorDB as PostgreSQL (pgvector)
    participant LLM as LLM Provider

    Student->>Gateway: POST /api/v1/ai/query {chapterId, question}
    Gateway->>Gateway: Validate User Session & Rate Limit
    Gateway->>RAG: Forward Query Request
    RAG->>VectorDB: Execute Vector Similarity Search (HNSW)
    VectorDB-->>RAG: Return Top Matching Passages + Page Numbers
    RAG->>RAG: Construct Guardrailed System Prompt
    RAG->>LLM: Stream Prompt Payload
    LLM-->>RAG: Stream Answer Tokens
    RAG-->>Student: SSE Stream Answer + Inline Citations
```

11. Technology Stack Specification

Component	Selected Technology	Role & Justification
Frontend SPA	React 18 & TypeScript	Component-based UI with compile-time type safety
Backend Framework	Spring Boot 3.x (Java 21)	Enterprise service execution and REST API management
Database Engine	PostgreSQL 16	Relational operational data storage with ACID guarantees
Vector Search	pgvector Extension	Native vector search integrated inside PostgreSQL
Object Storage	AWS S3 / MinIO	Scalable, durable storage for uploaded PDF study materials
AI / RAG Pipeline	Large Language Model	Passage embedding and grounded answer generation

DOCUMENT 3: DATABASE DESIGN DOCUMENT

1. Database Architecture Overview

The LearnPulse AI database is built on **PostgreSQL 16**, augmented by the native `pgvector` extension. This single-database architecture manages structured operational data (users, courses, quizzes) alongside high-dimensional vector embeddings within a unified, transactional storage engine.

2. Entity List & Domain Categorization

LEARNPULSE AI DATABASE ENTITIES			
AUTHENTICATION	LEARNING	CONTENT	ASSESSMENT
- users	- students	- notes	- quizzes
- refresh_tokens	- teachers	- documents	- quiz_questions
- audit_logs	- subjects	- doc_chunks	- question_options
	- chapters	- embeddings	- quiz_attempts
	- enrollments		- student_answers
AI & CONTEXTUAL RAG		ADMINISTRATION	
- ai_conversations		- teacher_subjects	
- ai_messages		- system_settings	
- retrieved_chunks		- student_progress	

3. Entity Descriptions

- **users**: Central account identity store for all platform users.
- **students**: Academic profile extensions for student users.
- **teachers**: Faculty profile extensions for instructors.
- **subjects**: Master record of academic courses.
- **chapters**: Sequential modules constituting a subject's syllabus.
- **notes**: Rich-text lecture notes published by instructors.
- **documents**: Metadata for uploaded PDF study files.
- **document_chunks**: Segmented text passages extracted from PDFs for semantic retrieval.
- **document_embeddings**: High-dimensional vector store enabled by `pgvector`.
- **quizzes**: Assessment configurations associated with course chapters.
- **quiz_questions**: Questions defining an assessment.
- **question_options**: Multiple-choice options for assessment questions.
- **quiz_attempts**: Records of student assessment executions and final grades.
- **student_answers**: Audit record of individual answers submitted during an attempt.

- `student_progress`: Reading and completion tracking per student per chapter.
 - `ai_conversations`: Context-bounded AI chat sessions.
 - `ai_messages`: Individual messages within an AI conversation.
 - `retrieved_chunks`: Audit table mapping AI messages to exact document citations.
-

4. Complete Entity Relationship Diagram (ERD)

```

erDiagram
    USERS ||--o| STUDENTS
    : "profile"
    USERS ||--o| TEACHERS : "profile"
    TEACHERS ||--o{ TEACHER_SUBJECTS : "teaches"
    SUBJECTS ||--o{ TEACHER_SUBJECTS : "managed by"
    SUBJECTS ||--o{ CHAPTERS : "contains"
    SUBJECTS ||--o{ ENROLLMENTS : "enrolled"
    STUDENTS ||--o{ ENROLLMENTS : "attends"
    CHAPTERS ||--o{ NOTES : "includes"
    CHAPTERS ||--o{ DOCUMENTS : "includes"
    CHAPTERS ||--o{ QUIZZES : "evaluates"
    DOCUMENTS ||--o{ DOCUMENT_CHUNKS : "parsed into"
    DOCUMENT_CHUNKS ||--o{ DOCUMENT_EMBEDDINGS : "indexed as"
    QUIZZES ||--o{ QUIZ_QUESTIONS : "contains"
    QUIZ_QUESTIONS ||--o{ QUESTION_OPTIONS : "has"
    STUDENTS ||--o{ QUIZ_ATTEMPTS : "takes"
    QUIZZES ||--o{ QUIZ_ATTEMPTS : "records"
    QUIZ_ATTEMPTS ||--o{ STUDENT_ANSWERS : "contains"
    QUIZ_QUESTIONS ||--o{ STUDENT_ANSWERS : "answers"
    STUDENTS ||--o{ AI_CONVERSATIONS : "starts"
    CHAPTERS ||--o{ AI_CONVERSATIONS : "bounded by"
    AI_CONVERSATIONS ||--o{ AI_MESSAGES : "contains"
    AI_MESSAGES ||--o{ RETRIEVED_CHUNKS : "cites"
    DOCUMENT_CHUNKS ||--o{ RETRIEVED_CHUNKS : "source"

```

```

    USERS {
        uuid id PK
        string email
        string first_name
        string last_name
        string role
    }
    SUBJECTS {
        uuid id PK
        string code
    }
    CHAPTERS {
        uuid id PK
        integer subject_id FK
        string title
        integer sequence_order
    }
    DOCUMENTS {
        uuid id PK
        uuid chapter_id FK
        string title
        string status
    }
    DOCUMENT_CHUNKS {
        uuid id PK
        uuid document_id FK
        text chunk_content
        integer page_number
    }
    QUIZZES {
        uuid id PK
        uuid chapter_id FK
        string title
        integer duration_minutes
    }

```

5. Core Database Schema Tables

5.1 Identity & User Domain Tables

Table: `users`

- **Purpose:** Primary identity record for all platform users.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Unique user identifier
email	VARCHAR(255)	NOT NULL, UNIQUE	User email address (login identifier)
password_hash	VARCHAR(255)	NOT NULL	Encrypted password hash
first_name	VARCHAR(100)	NOT NULL	Given name
last_name	VARCHAR(100)	NOT NULL	Surname
role	VARCHAR(20)	NOT NULL	Role: 'STUDENT', 'TEACHER', 'ADMIN'
is_active	BOOLEAN	NOT NULL	Account active toggle

Table: `students`

- **Purpose:** Extended academic profile for student users.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Profile ID
user_id	UUID	FOREIGN KEY -> users(id)	Linked user account
enrollment_number	VARCHAR(50)	NOT NULL, UNIQUE	Institutional student ID
academic_year	VARCHAR(20)	NOT NULL	Current academic year

Table: `teachers`

- **Purpose:** Extended profile for faculty members.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Profile ID
user_id	UUID	FOREIGN KEY -> users(id)	Linked user account
department	VARCHAR(100)	NOT NULL	Academic department
qualification	VARCHAR(100)	NULL	Faculty qualification

5.2 Course Content Domain Tables

Table: `subjects`

- **Purpose:** Master record of academic courses.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Subject ID
code	VARCHAR(20)	NOT NULL, UNIQUE	Course code (e.g. "CS-401")
title	VARCHAR(200)	NOT NULL	Course title
description	TEXT	NULL	Detailed course overview
is_active	BOOLEAN	NOT NULL	Subject availability toggle

Table: `chapters`

- **Purpose:** Sequential modules constituting a subject syllabus.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Chapter ID
subject_id	UUID	FOREIGN KEY -> subjects(id)	Parent subject
title	VARCHAR(200)	NOT NULL	Chapter header
sequence_order	INTEGER	NOT NULL	Position in syllabus

Table: `notes`

- **Purpose:** Rich-text lecture notes published by instructors.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Note ID
chapter_id	UUID	FOREIGN KEY -> chapters(id)	Parent chapter
title	VARCHAR(200)	NOT NULL	Note title
content_html	TEXT	NOT NULL	Formatted note content

Table: `documents`

- **Purpose:** Metadata for uploaded PDF study files.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Document ID
chapter_id	UUID	FOREIGN KEY -> chapters(id)	Parent chapter

title	VARCHAR(255)	NOT NULL	Original filename
file_key	VARCHAR(500)	NOT NULL	Object storage key
status	VARCHAR(20)	NOT NULL	Status: 'PENDING', 'READY', 'FAILED'

Table: `document\_chunks`

- **Purpose:** Segmented passages extracted from PDFs for semantic retrieval.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Chunk ID
document_id	UUID	FOREIGN KEY -> documents(id)	Parent document
chunk_index	INTEGER	NOT NULL	Passage index
chunk_content	TEXT	NOT NULL	Extracted passage text
page_number	INTEGER	NULL	Source page number

Table: `document\_embeddings`

- **Purpose:** High-dimensional vector store enabled by `pgvector`.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Embedding ID
chunk_id	UUID	FOREIGN KEY -> document_chunks(id)	Parent passage chunk
embedding	vector(1536)	NOT NULL	Semantic vector array

5.3 Assessment Domain Tables

Table: `quizzes`

- **Purpose:** Chapter assessment configurations.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Quiz ID
chapter_id	UUID	FOREIGN KEY -> chapters(id)	Parent chapter

title	VARCHAR(200)	NOT NULL	Quiz title
duration_minutes	INTEGER	NOT NULL	Allowed time limit
passing_score	INTEGER	NOT NULL	Pass threshold percentage

Table: `quiz\_questions`

- **Purpose:** Individual questions defining an assessment.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Question ID
quiz_id	UUID	FOREIGN KEY -> quizzes(id)	Parent quiz
question_text	TEXT	NOT NULL	Question statement
question_type	VARCHAR(20)	NOT NULL	Type: 'MCQ', 'TRUE_FALSE'
points	INTEGER	NOT NULL	Score value

Table: `question\_options`

- **Purpose:** Multiple-choice options for assessment questions.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Option ID
question_id	UUID	FOREIGN KEY -> quiz_questions(id)	Parent question
option_text	TEXT	NOT NULL	Display option text
is_correct	BOOLEAN	NOT NULL	Correct answer flag

Table: `quiz\_attempts`

- **Purpose:** Student assessment executions and final grades.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Attempt ID
quiz_id	UUID	FOREIGN KEY -> quizzes(id)	Parent quiz
student_id	UUID	FOREIGN KEY -> students(id)	Student taking attempt
score_percentage	NUMERIC(5,2)	NULL	Final calculated score
status	VARCHAR(20)	NOT NULL	Status: 'IN_PROGRESS', 'SUBMITTED'

5.4 AI & Context Domain Tables

Table: `ai\_conversations`

- **Purpose:** Context-bounded AI study chat sessions.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Conversation ID
student_id	UUID	FOREIGN KEY -> students(id)	Owning student
chapter_id	UUID	FOREIGN KEY -> chapters(id)	Bounding course chapter
title	VARCHAR(200)	NOT NULL	Session title

Table: `ai\_messages`

- **Purpose:** Individual messages within an AI conversation.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Message ID
conversation_id	UUID	FOREIGN KEY -> ai_conversations(id)	Parent session
sender_type	VARCHAR(10)	NOT NULL	Sender: 'USER', 'ASSISTANT'
message_text	TEXT	NOT NULL	Message body

Table: `retrieved\_chunks`

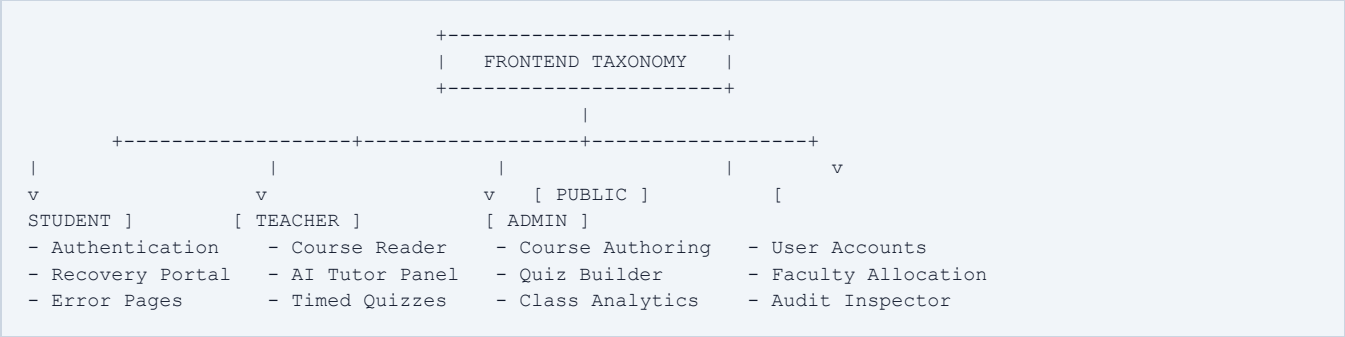
- **Purpose:** Citation map linking AI answers to retrieved source passages.

Column	Data Type	Key / Constraints	Description
id	UUID	PRIMARY KEY	Map ID
message_id	UUID	FOREIGN KEY -> ai_messages(id)	AI assistant message
chunk_id	UUID	FOREIGN KEY -> document_chunks(id)	Retrieved source passage
similarity_score	NUMERIC(5,4)	NOT NULL	Vector similarity score

DOCUMENT 4: FRONTEND DESIGN DOCUMENT

1. Information Architecture Strategy

The LearnPulse AI frontend is structured as a responsive **React 18 Single Page Application (SPA)**. It enforces a strict role-based layout taxonomy, providing tailored workspace layouts for Students, Teachers, and Administrators.



2. Page Taxonomy & Route Maps

2.1 Public Unauthenticated Routes

- /login: Primary user authentication portal.
- /forgot-password: Password recovery request page.
- /404: Page not found boundary.

2.2 Student Workspace Routes (Wrapped in `StudentLayout`)

- /student/dashboard: Enrolled subjects overview, upcoming quizzes, study activity.
- /student/subjects: Directory of active enrolled courses.
- /student/subjects/:subjectId/chapters/:chapterId: Main chapter learning reader featuring split-screen Notes, PDF Viewer, and expandable AI Tutor drawer.
- /student/quizzes/:quizId/take: Interactive quiz attempt workspace with countdown timer.
- /student/quizzes/:quizId/results/:attemptId: Quiz results breakdown and answer explanations.

2.3 Teacher Workspace Routes (Wrapped in `TeacherLayout`)

- /teacher/dashboard: Overview of assigned subjects, document ingestion statuses, class alerts.
- /teacher/subjects/:subjectId/chapters: Chapter module reordering and syllabus builder.
- /teacher/subjects/:subjectId/chapters/:chapterId/edit: Rich-text note editor and PDF upload interface.
- /teacher/quizzes/create: Quiz authoring tool and question builder.
- /teacher/analytics: Class performance distribution, topic difficulty metrics, and top AI query trends.

2.4 Administration Workspace Routes (Wrapped in `AdminLayout`)

- /admin/dashboard: System health overview, database storage metrics, vector search latencies.

- /admin/users: User account management directory (create, status toggle, role edit).
- /admin/subject-assignments: Interface for allocating faculty to academic subjects.
- /admin/audit-logs: System audit trail inspector.

3. Role-Based Access Control & Layout Guards

```
[ Incoming URL Request ]           |           v
< Session Active? >  --- No ---> Redirect to /login
|                               v < Role Authorized?
> --- No ---> Redirect to /403 Forbidden
|
Yes
v
[ Render Role Layout Component ]
```

4. User Navigation Flows

4.1 Student Navigation Flow

```
flowchart LR
    Login[Login Page] --> StudentDash[Student Dashboard]
    StudentDash --> SubjectView[Subject Chapter List]
    SubjectView --> ChapterReader[Chapter Content Reader]
    ChapterReader <--> |Toggle Drawer| AITutor[AI Tutor Chat Drawer]
    SubjectView --> QuizList[Quiz Directory]
    QuizList --> QuizTake[Take Quiz Workspace]
    QuizTake --> QuizResults[Quiz Score & Breakdown]
```

4.2 Teacher Navigation Flow

```
flowchart LR
    Login[Login Page] --> TeacherDash[Teacher Dashboard]
    TeacherDash --> SyllabusMgmt[Subject & Chapter Builder]
    SyllabusMgmt --> NoteEditor[Rich-Text Note Editor]
    SyllabusMgmt --> PDFUpload[PDF Document Uploader]
    TeacherDash --> QuizBuilder[Quiz Authoring Workspace]
    TeacherDash --> AnalyticsView[Class Performance Analytics]
```

4.3 Administrator Navigation Flow

```
flowchart LR
    Login[Login Page] --> AdminDash[Admin Console]
    AdminDash --> UserMgmt[User Management Table]
    AdminDash --> FacultyMap[Teacher-Subject Mapping]
    AdminDash --> AuditLog[Security Audit Trail Viewer]
```

5. Dashboard Structures

Student Dashboard

- **Header:** Welcome message, academic year, personal progress summary.
- **Main Area:** Enrolled course cards showing chapter completion progress bars.
- **Sidebar:** Pending quizzes, recent AI Q&A; sessions, quick-start study buttons.

Teacher Dashboard

- **Header:** Assigned subjects overview, total student count.
- **Main Area:** Course management cards, recent PDF processing statuses ("Ready" / "Processing").
- **Sidebar:** Quiz submission activity, class grade averages, AI query topic trends.

6. UI Module Boundaries & Component Hierarchy

```
[ Application Root ]
  ■■■ [ AuthProvider ] (Manages JWT & User Session)
  ■■■ [ Router ]
    ■ ■■■ [ PublicLayout ]
      ■ ■■■ LoginView / ForgotPasswordView
      ■ ■■■ [ StudentLayout ]
        ■ ■■■ NavigationSidebar
        ■ ■■■ ChapterReaderView
        ■ ■ ■ ■■■ RichTextNoteViewer
        ■ ■ ■ ■■■ PDFDocumentViewer
        ■ ■ ■ ■■■ AITutorDrawer (SSE Client + Citation Links)
        ■ ■ ■ ■■■ QuizExecutionView (Timer + Question Panel)
        ■ ■■■ [ TeacherLayout ]
          ■ ■■■ CourseManagementView
          ■ ■■■ PDFUploadModal (Progress + Status Badge)
          ■ ■■■ QuizAuthoringView
        ■ ■■■ [ AdminLayout ]
          ■ ■■■ UserManagementTable / AuditLogViewer
```

DOCUMENT 5: AI SYSTEM DESIGN DOCUMENT

- **Document Title:** AI Subsystem & Contextual RAG Architecture — LearnPulse AI
- **Document Version:** 1.0.0-FINAL
- **Status:** Approved for AI Pipeline Implementation
- **Target Audience:** AI Engineers, Backend Engineers, Solution Architects

1. AI Subsystem Goals & Vision

The core mission of the LearnPulse AI subsystem is to provide an **intelligent, grounded digital academic tutor**.

IMPORTANT:

****The Zero-Hallucination Imperative**:** The AI Tutor operates under strict context bounding. Answers are generated exclusively from document passages uploaded by course instructors. If a student query cannot be answered using the retrieved course context, the AI gracefully declines to answer, preventing hallucinated or off-syllabus responses.

2. Contextual RAG Paradigm Overview

LearnPulse AI uses **Retrieval-Augmented Generation (RAG)** to connect Large Language Models with institutional course documents.

```
[ Teacher Uploads PDF ] ---> [ Text Parsing ] ---> [ Passage Chunking ] ---> [ Generate Vector Embeddings ]
---> [ Save to pgvector ]

[ Student Query ] -----> [ Query Vector ] ----> [ Similarity Search ] -> [ Extract Top Passages ] -----
---> [ Grounded Answer + Citations ]
```

3. Document Ingestion & Parsing Workflow

1. **Upload Trigger:** Instructor uploads a PDF document associated with a specific chapter.
2. **Text Extraction:** The document service parses text while tracking source page numbers.
3. **Passage Chunking:** Clean text is split into logical passages for semantic indexing.
4. **Vector Embedding:** Each passage is transformed into a high-dimensional vector using an embedding model.
5. **Database Storage:** Text passages and vectors are stored in PostgreSQL using `pgvector`.

4. Chunking Strategy

- **Strategy:** Token-aware recursive character splitting.
- **Target Passage Size:** ~500 words (~2,000 characters).
- **Passage Overlap:** 50 words (~200 characters) between consecutive passages to preserve context across boundaries.
- **Metadata Tracking:** Every passage retains references to `document_id`, `chapter_id`, and `source page_number`.

5. Embedding Strategy & Vector Storage

- **Embedding Model:** High-accuracy semantic embedding model converting passages into 1536-dimensional float arrays.
- **Vector Engine:** PostgreSQL 16 with native `pgvector` extension.
- **Indexing Mechanism:** Hierarchical Navigable Small World (HNSW) index using cosine distance operators for fast vector similarity searches.

6. Retrieval Pipeline & Cosine Similarity Search

When a student submits a question within a chapter:

1. The question string is converted into a query vector.
2. The system executes a vector similarity query in `pgvector`, filtered strictly by `chapter_id`.
3. The top matching passages meeting a minimum similarity threshold (≥ 0.70) are retrieved for prompt context assembly.

7. Context Assembly & Prompt Construction

The retrieved passages are formatted into a structured system prompt before being sent to the LLM.

```
+-----+
|                                     |
|               STRUCTURED RAG SYSTEM PROMPT               |
|                                     |
+-----+
| ROLE: You are the official LearnPulse AI Academic Tutor for this course. |
|                                     |
| INSTRUCTIONS:                                              |
| 1. Answer the student's question ONLY using the verified context passages below. |
| 2. For every statement, include an inline citation: [Document Title, Page X]. |
| 3. If the context does not contain the answer, reply:      |
|    "I am sorry, but this topic is not covered in your chapter study materials." |
| 4. Do NOT use outside knowledge or answer off-topic questions. |
|                                     |
| VERIFIED COURSE CONTEXT PASSAGES:                          |
| - [Doc: Operating_Systems.pdf, Page 14]: "Virtual memory decouples logical..." |
| - [Doc: Operating_Systems.pdf, Page 15]: "Paging divides physical memory..." |
|                                     |
| STUDENT QUESTION: "Explain how virtual memory works."     |
+-----+
```

8. Streamed AI Response Generation & Citation Mapping

- **Streaming Mechanism:** Answers stream to the client UI in real time using Server-Sent Events (SSE) for low latency.
- **Citation Persistence:** Source passage IDs and similarity scores are saved to the `retrieved_chunks` table, creating an audit trail for every generated response.

9. End-to-End RAG Sequence Diagram

```
sequenceDiagram
    actor Student
    participant Gateway as Student Browser
    participant RAG as API Gateway
    participant VectorDB as RAG Engine (pgvector)
    participant LLM as VectorDB as PostgreSQL
    participant LLM as LLM Provider
    participant LLM as LLM Service

    Student->>Gateway: Submit Question (Chapter ID + Query)
    Gateway->>RAG: Forward Validated Request
    RAG->>VectorDB: Similarity Search (Query Vector vs Chapter Passages)
    VectorDB-->>RAG: Return Top Matching Passages + Page Numbers
    RAG->>RAG: Assemble Guardrailed System Prompt
    RAG->>LLM: Send Structured Prompt Payload
    LLM-->>RAG: Stream Generated Answer Tokens
    RAG-->>Student: SSE Streamed Response with Page Citations
```

10. AI Guardrails & Prompt Injection Defense

SECURITY POLICY:

Safety Guardrails:

Input Sanitization: User inputs are stripped of prompt override strings (e.g., "Ignore previous instructions").

Context Isolation: System prompts mandate that answers must be derived solely from retrieved passages.

Audit Logging: Flagged prompt injection attempts write security alerts to the `audit_logs` table for administrative review.

11. AI Limitations & Known Boundaries

- Text-Only Ingestion (Phase 1):** Ingestion extracts textual content from PDFs (embedded images/diagrams are skipped in Phase 1).
- Course Boundary:** The AI cannot answer questions spanning non-enrolled subjects or general topics outside uploaded course documents.

12. Future AI Enhancements & Roadmap

- Phase 2 — Multi-Modal Ingestion:** Parsing mathematical formulas (LaTeX), diagrams, and video/audio lecture transcripts.

- **Phase 3 — AI-Assisted Quiz Drafting:** Automated draft question generation from uploaded PDFs for instructor review.
 - **Phase 4 — Adaptive Learning Recommendations:** Suggesting specific chapter reading passages based on student quiz mistakes.
-

End of LearnPulse AI Master Engineering Design Package.